

Software Engineering

Books or notes are **not** allowed.

Write only on these sheets. **Concise** and **readable** answers please.

Surname, name, matricola _____

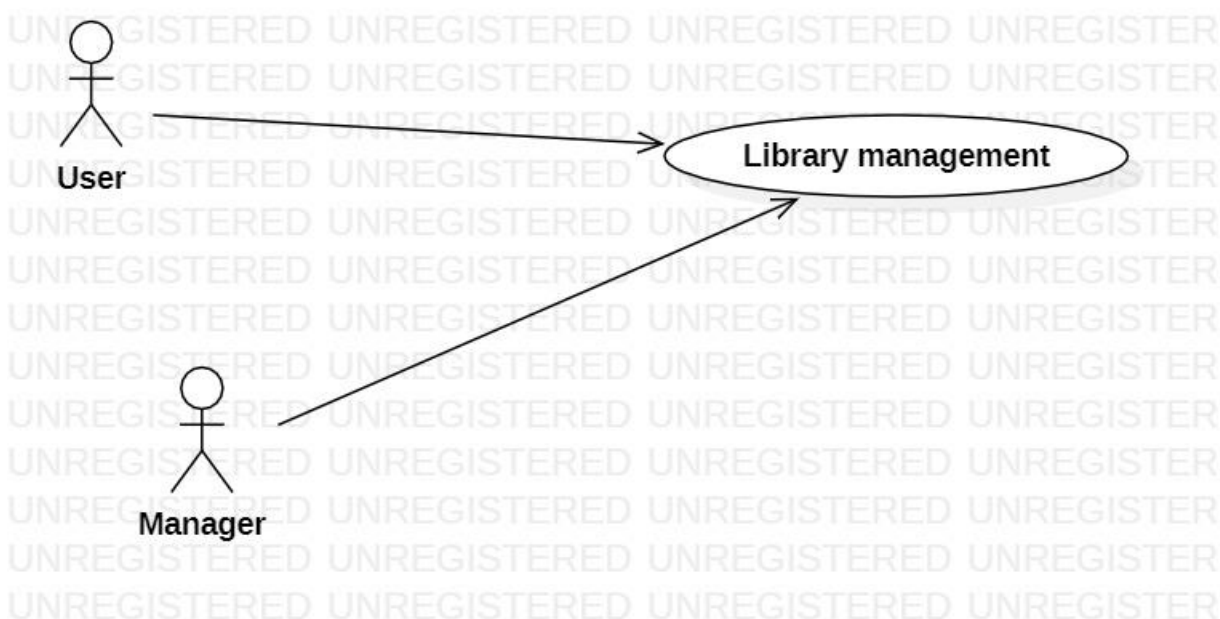
Library management

A library, managing physical or digital books, is supported by a web application. The application must support browsing the catalogue of books, managing book loans (record the start and end of loan, send notifications about loan expiration and similar), reserving books not available (already borrowed by other users), requesting to buy books not in the library, defining and managing accounts for users, download book (in case of digital books). Digital books are downloaded as pdf or epub files protected by DRM (Digital Right Management). On the user side, the user can read digital books only through a specific app, to be downloaded and installed on a smartphone.

1 – a. Define the **context diagram** (including relevant interfaces)

Actor	Physical interface	Logical interface
User	PC / smartphone	GUI
Library manager	PC	GUI

In some cases DRM check requires to access an external service, that should be represented as an actor too. The ebook reader is considered inside the system, so it is not represented as an actor.

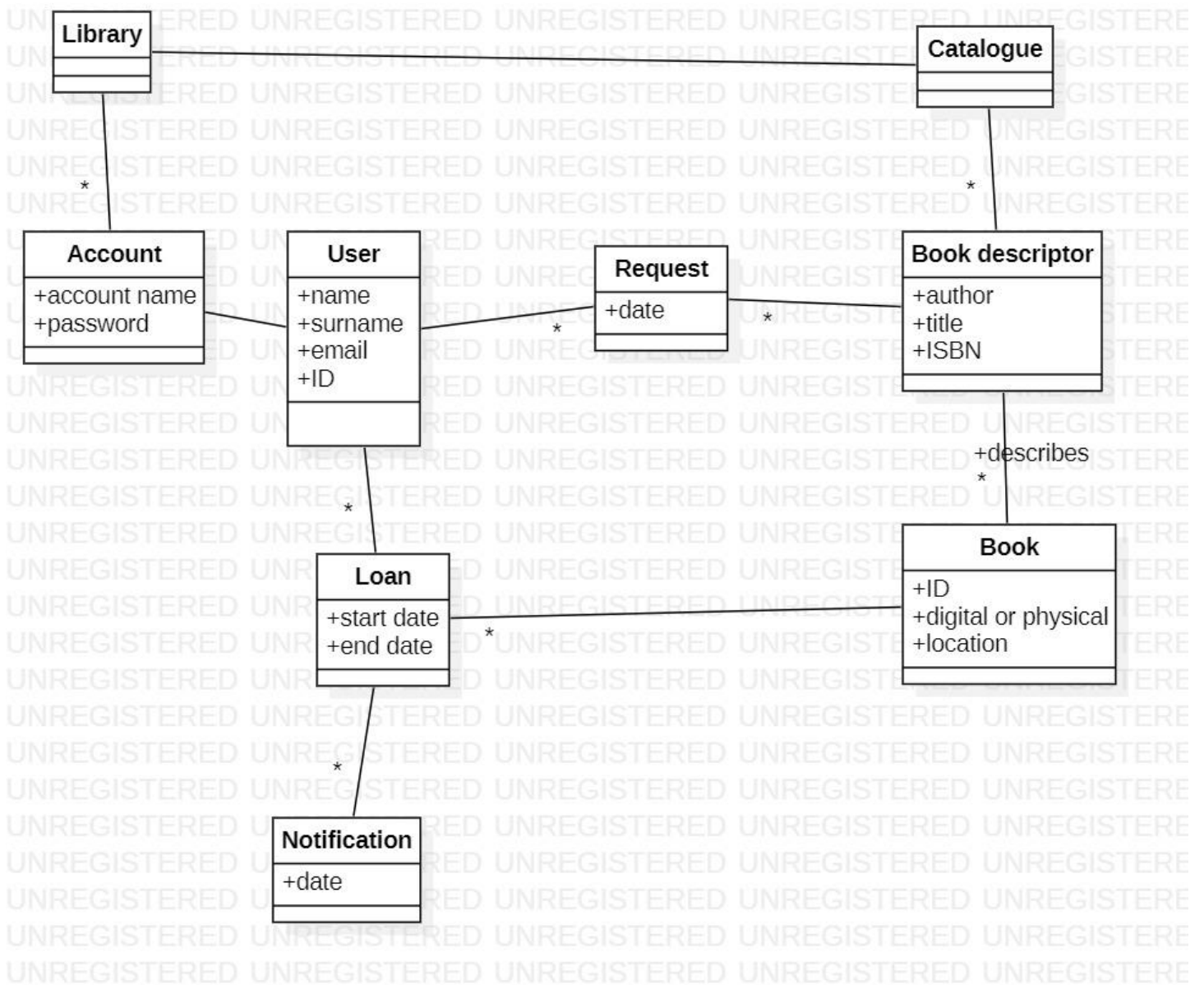


1-b Define the **glossary** (key concepts and their relationships) (UML class diagram) for the application

User and account could be merged in one class.

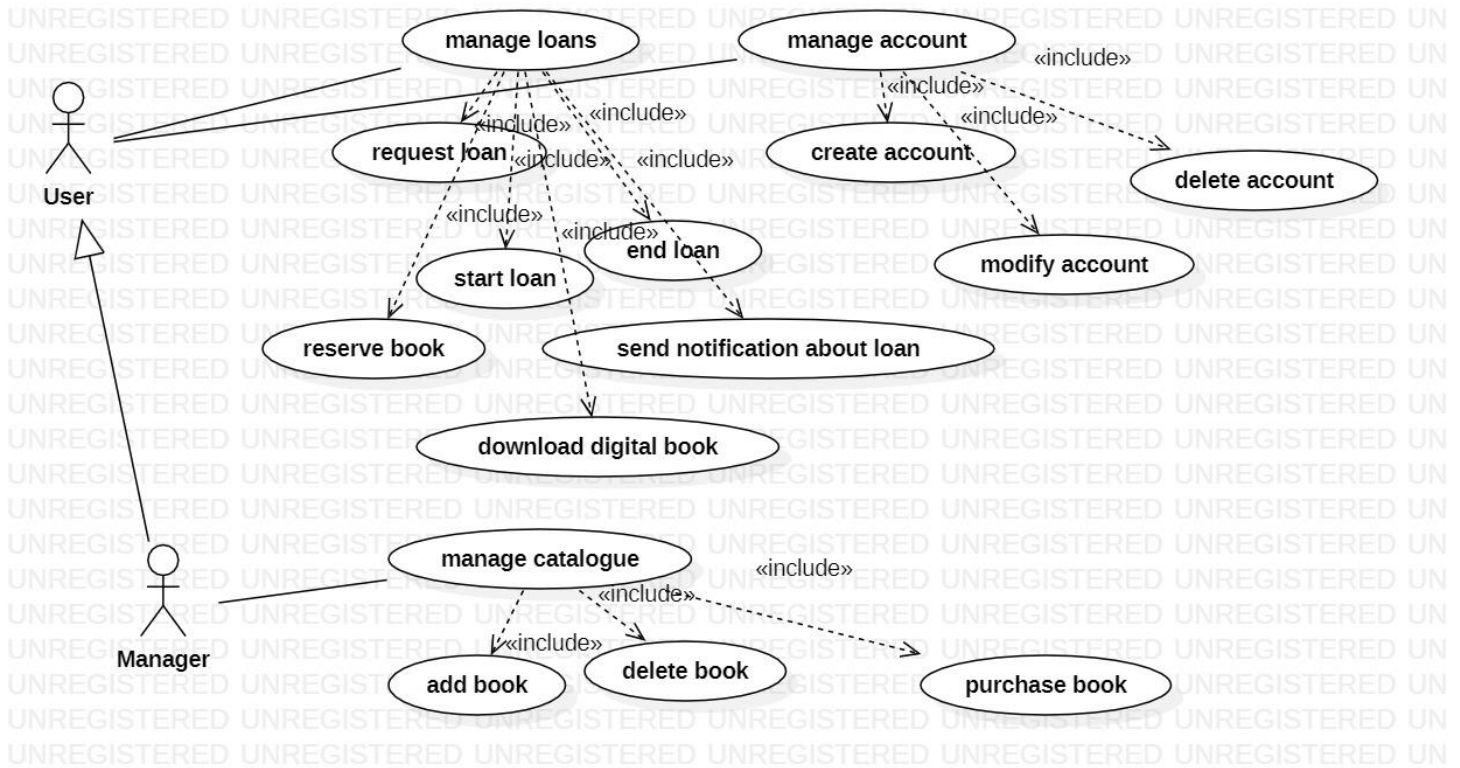
Book descriptor and book instance follow a known pattern

Book models both digital and paper books, another option could be to define two sub classes

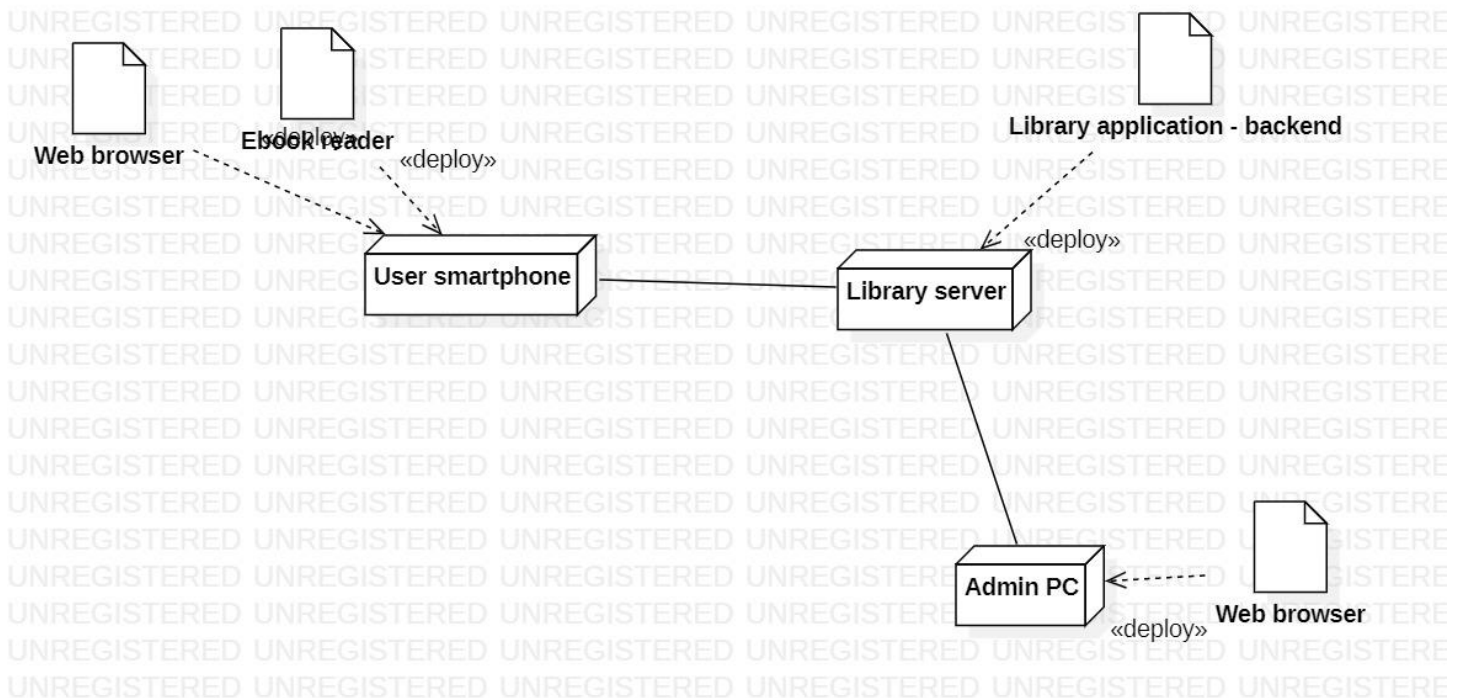


1-c Draw the Use Case Diagram for the application. For each Use Case give a short description here

Use Case ID	Description



1-d Draw the deployment diagram



2 Define black box tests for the following function, using equivalence classes and boundary conditions.

The function receives an array of number and returns the most frequent number in the array, or undefined if the array is empty.

In case more numbers have the same frequency, it returns the number that appears first in the array

```
function findFrequency(numbers: number[]): number | undefined
```

ex [1,2,3,4,5,4] → 4

[1,2,3,4,5] → 1

[] → undefined

Criteria: value of number

Number of elements in array numbers: 0,1, many

Elements repeated, yes or not

Number	Number of elements	Elements repeated	Test cases
[minint, maxint]	0	-	T1 [] → undefined
	1	-	T2 [10] → 10 T2B1[minint-1] → undefined T2B2[minint] T2B3[minint+1] T2B4[maxint -1] T2B5[maxint] T2B6[maxint +1]
-	Many	N	T3 [4,7,1] → 4
-	-	Y	T4 [4,7,7,1] → 7 T5 [4,7,7,1,1] → 7

Boundary conditions.

Number: minint -1, minint, minint +1, maxint-1, maxint, maxint +1

Number of elements : 0,1, maxint

3 For the function above and the implementation below, define test cases to obtain the highest possible node coverage, edge coverage, multiple condition coverage, loop coverage, path coverage.
For the test cases, **write only the input value**.

```

1. function findFrequency (numbers: number[]): number | undefined {
2.     if (numbers.length === 0) return undefined;
3.
4.     const frequencyMap: { [key: number]: number } = {};
5.     let maxFrequency = 0;
6.     let mode = numbers[0];
7.
8.     for (const num of numbers) {
9.         if (frequencyMap[num]) {
10.             frequencyMap[num]++;
11.         } else {
12.             frequencyMap[num] = 1;
13.         }
14.
15.         if (frequencyMap[num] > maxFrequency) {
16.             maxFrequency = frequencyMap[num];
17.             mode = num;
18.         }
19.     }
20.
21.     return mode;
22.}

```

Coverage type	Number of test cases needed to obtain 100% coverage	Coverage obtained with test cases defined (%)	Test cases defined
statement	3 (or even 2)	100	T1 T4
decision	3 (or even 2)	100	T1 T4
Loop	3	Zero loop non feasible, 66	T2 one loop T4 many loops
Path	Order of $4^{\text{number.length}}$	Close to zero	

4 Which of the following statements correctly expresses a Usability non functional requirement?

- A. All functions should be easy to use
- B. The application should be very easy to use by all user
- C. Any user who uses a computer since more than 2 years should be able to use function X
- D. Out of ten users, at least 5 should rate the application as 'very usable' (on a 3 levels scale, non usable, usable, very usable)
- E. The application should be easy to learn by first time users.

5 Interactions and conflicts between non functional requirements

- A. Never happen, if they happen it depends on ill written or wrong NF requirements
- B. Can happen, and can be accepted
- C. Can happen, and should be fixed, the developers can change the NF requirements as needed to avoid the conflict
- D. Can happen, and should be fixed, the end users should change the NF requirements as needed to avoid the conflict

6 A software application is developed without a requirements document. Testing the application can be accomplished through

- A. Inventing test cases on the fly after the application is coded
- B. Avoiding the testing phase
- C. Asking the end users to use the application directly in operation and fixing it if needed
- D. Testing is not feasible. A requirement document should be produced anyway, even after coding.
- E. Asking the end users to write the test cases

7 The estimate to develop an application is 1600 person hours. Development is made by a team of 4 people, working each 40 hours per week. The corresponding calendar duration could be

- A. More than 10 calendar weeks
- B. Less than 10 calendar weeks
- C. Exactly 10 calendar weeks
- D. More than 2.5 calendar months
- E. More than 10 calendar months

$1600 / (4 * 40) = 10$ calendar weeks. However, it is impossible (due to sick leaves, vacations, other engagements of the personnel, national holidays and similar) to end the project exactly in 10 weeks, it will be definitely more. Considering 4 weeks in a month, 10 weeks correspond roughly to 2.5 calendar months.